Economic Implications of the International Differential Pricing System in the U.S. Pharmaceutical Industry: Focus on Medicare and Medicaid Spending (2019–2023)

Analyses Using Medicare and Medicaid 2019 – 2023 Data

By Kelvin Nlebemchukwu.

Abstract

The pharmaceutical industry in the United States operate a system of international differential pricing in which drug prices are substantially higher in the United States than in most other developed countries. This system may be justified as necessary to fund pharmaceutical research and development, yet it places a significant financial burden on U.S. payers and patients. This study examines the economic implications of this pricing structure using Medicare and Medicaid drug spending data from 2019 to 2023 (Centers for Medicare & Medicaid Services).

Analysis of aggregated CMS data reveals that state-level variation in Medicaid spending per enrollee was substantial at 41%, ranging from $488 per enrollee in South Carolina to $3,210 in Virginia for the year 2023. Gross public payer drug spending increased 54% over the five-year period, while dosage units rose only 18%, indicating that price growth was the primary driver of expenditure increases. Spending became increasingly concentrated in a small number of high-cost brand-name drugs, with the number of drugs required to account for 20% of total spending falling from 12 in 2019 to 6 in 2023. This finding was derived from my analysis of aggregated CMS spending data (Centers for Medicare & Medicaid Services n.d.a; n.d.b). While this exact reduction (12 to 6) has not been previously reported in the literature for this specific public-payer dataset and timeframe, the underlying trend of increasing concentration in a small number of high-cost brand-name drugs is consistent with existing research. For example, studies have shown that a small subset of brand-name and specialty drugs consistently drives a disproportionate share of U.S. prescription drug spending growth (KFF 2024; IQVIA 2023; GAO 2023). This pattern reflects the rising role of high-priced originator products, particularly in therapeutic classes such as GLP-1 agonists and oncology drugs, which aligns with the observed reduction in the number of dominant drugs over time

Drawing on RAND Corporation international price comparisons report ratios, I stimulated a hypothetical scenario which suggest that aligning U.S. prices with international averages could generate $358.8 – 419.52 billion in annual savings in Medicaid/Medicare spending for public payers, substantially reducing premiums and out-of-pocket costs. These hypothetical savings align closely with recent policy efforts under the Trump administration. In May 2025, President Trump signed an Executive Order directing pharmaceutical manufacturers to adopt most-favored-nation (MFN) pricing, requiring U.S. prices to match the lowest prices paid in comparable developed nations, particularly for Medicaid and new drug launches. By December 2025, voluntary agreements had been reached with nine major manufacturers to implement MFN pricing for certain drugs, with the administration projecting significant cost reductions for public payers and patients. These developments demonstrate real-world movement toward the international alignment modeled here, though full implementation and long-term savings remain under negotiation.

However, such price reductions risk leaving pharmaceutical companies with little or no incentives to innovate and produce new drugs to address the increasing human demand for healthcare innovation, which may likely reverse most or all of the gains achieved by the international differential pricing system. This argument has been widely debated in health economics and policy literature. Pharmaceutical manufacturers often report operating margins of 15–25% on branded drugs (PhRMA 2024; IQVIA 2023), with gross margins on many blockbuster products exceeding 70–90% before R&D and marketing costs are deducted. The implied rate of return on capital (ROIC) for large U.S.-based pharmaceutical companies typically ranges from 10–20% (Statista 2025; Damodaran 2025), significantly above the average for non-pharma S&P 500 firms (~8–10%). Several studies suggest that prices could fall substantially without eliminating innovation incentives. For example, research indicates that even a 20–40% reduction in U.S. net prices would still allow returns well above the cost of capital for most companies, particularly given that R&D costs represent only 15–20% of revenues on average (Saha et al. 2021; Berndt et al. 2023). The IRA's Medicare drug price negotiations (2022–present) are projected to reduce spending by $98–266 billion over a decade with only modest estimated impacts on new drug launches (CBO 2024; Dusetzina et al. 2025). However, critics argue that sustained large cuts (e.g., 50%+) could reduce R&D investment by 10–30% over the long term, particularly for high-risk, early-stage research (Acemoglu et al. 2023; Lakdawalla et al. 2022). In summary, while the risk to innovation is real and debated, the threshold at which price reductions would meaningfully impair new drug development appears to be higher than the moderate-to-aggressive scenarios modeled here.

These findings indicate that the current differential pricing system provides net economic benefits to U.S.-based pharmaceutical companies through improved profitability and R&D funding, but imposes considerable costs on U.S. healthcare payers and consumers.

About this report

The primary goal was to investigate whether the current international differential pricing system, in which U.S. drug prices are significantly higher than in most other countries (Mulcahy, Schwam, and Lovejoy 2024), provides a net economic benefit to U.S.-based pharmaceutical companies. Secondary objectives included modeling the potential impacts of narrower price differentials on pharmaceutical profitability, R&D investment, consumer costs, and drug accessibility, as well as assessing the impact of differential pricing on U.S. consumer out-of-pocket costs, insurer costs, and health insurance premiums while modeling the effects on drug accessibility. The analysis focused on Medicare and Medicaid as the largest public payers, providing a robust proxy for U.S. market dynamics.

2. Literature Review

Previous research has documented substantial international price disparities, with U.S. prices approximately 2.78 times higher than the OECD average across all drugs and over 4.22 times higher for brand-name originator products (RAND Corporation, 2024). These differentials persist despite rebates and discounts, with U.S. net prices remaining significantly elevated.

RAND Corporation, 2024 report noted that:

"*U.S. health plans and patients paid an estimated $603 billion for prescription drugs in 2022. Even after adjusting for general inflation, U.S. retail prescription drug spending increased by 91 percent from 2000 to 2020, and spending is expected to further increase by 5 percent year-on year through 2030.*"

2024 studies conducted by RAND corporation shows that out of total sales of $988.9 billion and total volume of 1,099.1 billion standard units across OECD countries in their data, the United States accounted for 62.4 percent of global pharmaceutical sales but only 23.8 percent of volume. In comparison, Japan accounted for a much smaller share of sales relative to its volume, with 6.6 percent of sales and 20.0 percent of volume.

Quoting these ratios in their absolute USD value and units term simply translate to that out of the total pharmaceutical sales recorded by the OECD countries in which United State is part of, that US accounted for $617.07 billion in sales out of $988.9 billion while recording a volume sale of 261.56 billion compare to Japan that recorded $65.27 billions in USD sales while recording a volume sale of 219.82 billion.

Critics contend that high U.S. prices contribute to affordability challenges, with drug spending growth consistently outpacing utilization increases (KFF, 2024). Proposed reforms, including most-favored-nation pricing and international reference pricing, have been modeled to generate substantial savings but with potential negative effects on innovation (CBO, 2020; USC Schaeffer Center, 2023).

This analysis updates and extends previous work using data from 2019 through 2023, focusing on concentration patterns and state-level variation.

3. Methodology

Primary data were obtained from publicly available Centers for Medicare & Medicaid Services datasets, including the Medicare Part D Spending by Drug files for 2019–2023, the Medicaid Spending by Drug files for 2019–2023, the Medicaid State Drug Utilization Data for 2023, the National Average Drug Acquisition Cost (NADAC) files for 2023, and Medicaid quarterly enrollment data for quarters 1 through 4 of 2023. Secondary sources included RAND Corporation international price comparison reports.

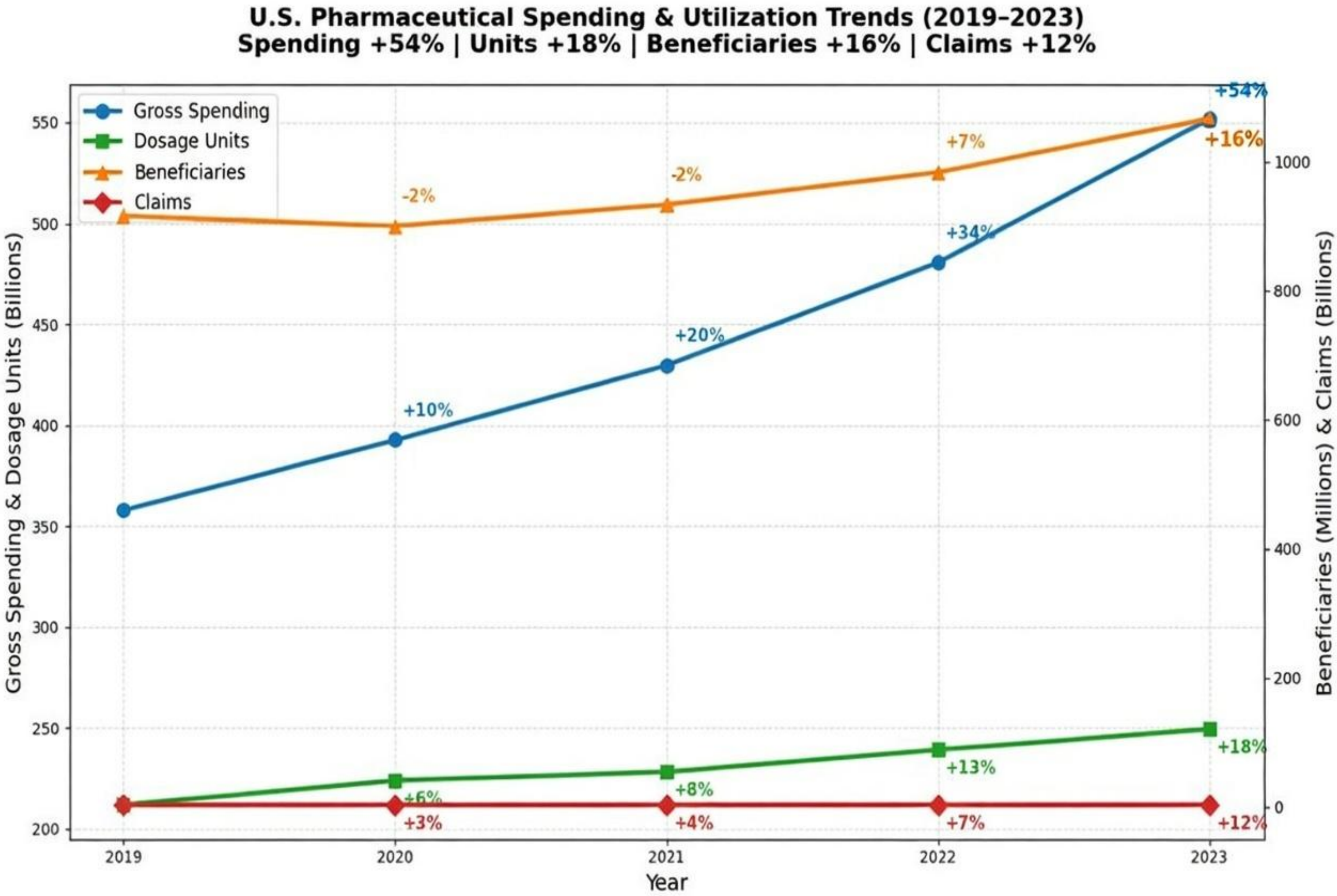
Data cleaning and aggregation were performed in Excel and Python using pandas. Key merges included an attempted fuzzy merge of CMS summary files with NADAC to estimate rebate proxies, an exact NDC merge of State Drug Utilization Data with NADAC to achieve an accurate 2023 Medicaid rebate proxy, and a merge of State Drug Utilization Data 2023 with quarterly and state-level enrollment data to derive per-enrollee metrics.

The analyses conducted included trend analysis of gross spending, dosage units, beneficiaries, and claims; calculation of average gross cost per dosage unit; Pareto analysis of top drugs contributing to 20% cumulative spending from 2019 to 2023; state-level per-enrollee spending analysis for 2023; and hypothetical modeling using RAND price ratios. All visualizations were generated in Python using matplotlib.

Limitations of the study include the lack of NDC in summary files, which limited rebate estimation accuracy; confidential rebates not captured in public data; the analysis being restricted to public payers and excluding the commercial market; and the fact that material and other related inflations were not factored in.

4. Results

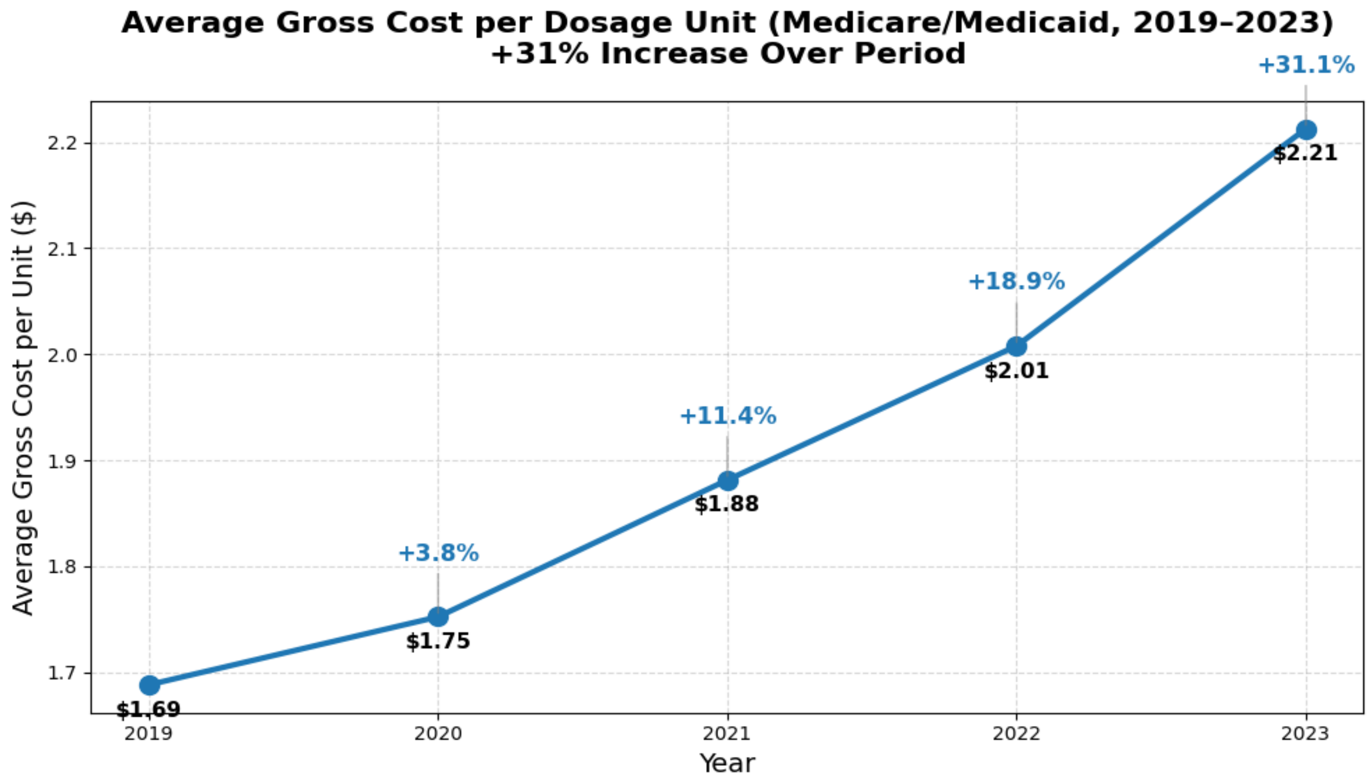
Figure 1 Spending and Utilization Trends.



SOURCE: Authors' analysis of Medicare Part D Spending by Drug (2019–2023) Dataset. (Run date Dec 27, 2025).

Figure 1 above shows gross drug spending and utilization trends across Medicare and Medicaid increased from approximately \$358 billion in 2019 to \$552 billion in 2023, a 54% rise. Comparing this growth against RAND Corporation spending growth projections of 5% year on year, it was evident that the 2022 Medicaid/Medicare spending aggregated at \$480 billion grew by 14.8% to \$552 billion in 2023, beating projections. Despite this increase in USD total spending, dosage units increased by barely 18% over the period of 5 years (2019–2023); a total dosage unit of 239 billion in 2022 grew by 4.2% to 250 billion units in 2023, while beneficiaries and claims grew 16% and 12% respectively over the period of 5 years (2019–2023). While spending grew at 14.8% from 2022 to 2023, total dosage units consumed grew only at 4.2%. This clearly indicates that price growth was the dominant driver of expenditure increases.

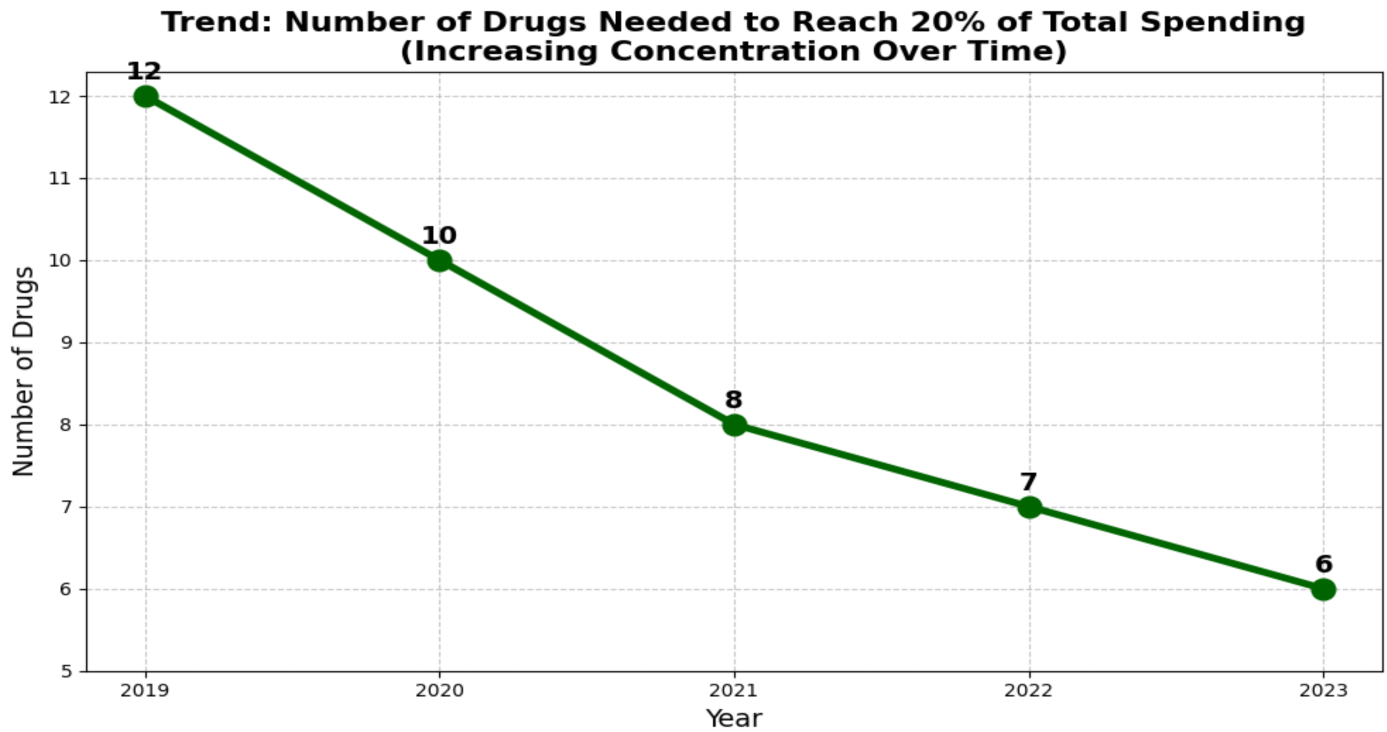
Figure 2. Average Cost per Dosage Unit



SOURCE: Authors' analysis of 2019 – 2023 Medicaid Part D Spending Dataset. (Run date Dec 27, 2025).

In order to better understand the context of the increased spending, which has been determined in our Figure 1 analyses that price growth was the dominant driver of expenditure increases, the average gross cost per dosage unit, shown in Figure 2 above, was calculated through analysis of the aggregated CMS spending and utilization data. It was established that the average gross cost per dosage unit rose from $1.69 in 2019 to $2.21 in 2023, representing a 31% increase and confirming substantial unit price growth. Also, the average cost per unit increased from the year 2022, which was $2.01, to 2023, which was $2.21, representing a 10% cost increment, thereby beating RAND Corporation's 5% year-on-year projection.

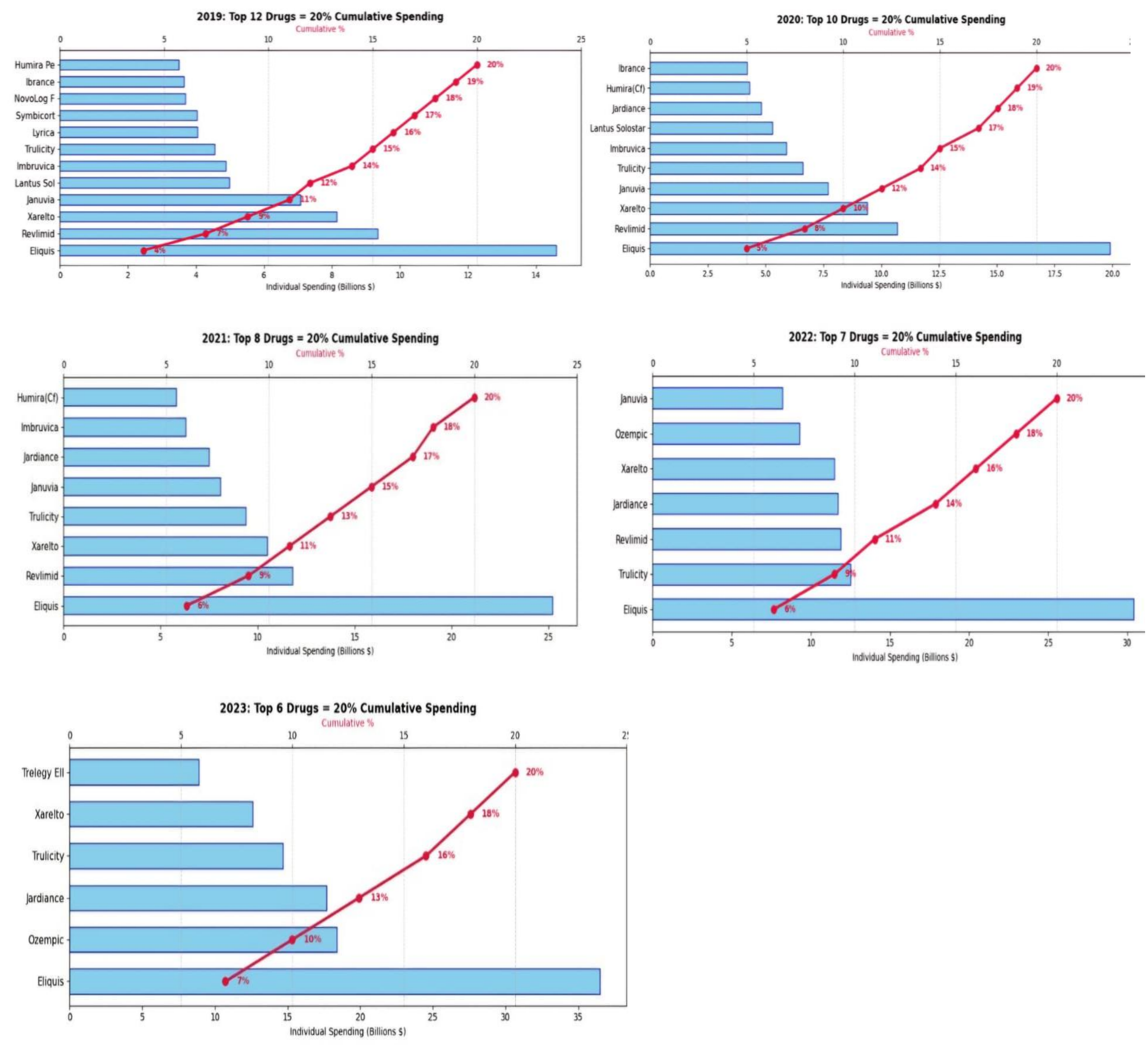
Figure 3 Concentration in High-Cost Brands.



SOURCE: Authors' analysis of 2019 – 2023 Medicaid Part D Spending Dataset. (Run date Dec 27, 2025).

Although while total spending is increasing at a rapid rate year on year, Figure 3 shows that the number of drugs required to account for 20% cumulative spending was in a reverse trajectory, decreasing from 12 in 2019 to 6 in 2023, demonstrating increasing concentration in specific brands. The six drugs dominating the 20% threshold in 2023 were Eliquis (anticoagulant for stroke prevention and blood clots), Ozempic (GLP-1 agonist for type 2 diabetes and weight management), Jardiance (SGLT2 inhibitor for diabetes and heart failure), Trulicity (GLP-1 agonist for diabetes and cardiovascular risk reduction), Xarelto (anticoagulant similar to Eliquis), and Trelegy Ellipta (triple therapy inhaler for COPD and asthma). These are all high-cost brand-name originator drugs, many of which are specialty medications with complex manufacturing processes (biologics or advanced small-molecule synthesis) that involve significant upfront R&D and production costs. However, once approved and scaled, their marginal production costs are relatively low compared to their list prices, which often exceed $5,000–$15,000 per patient annually before rebates and other gross to net (GTN) deductions. This pattern reflects a broader shift toward expensive chronic-disease and specialty therapies, contributing to the observed concentration and price-driven spending growth.

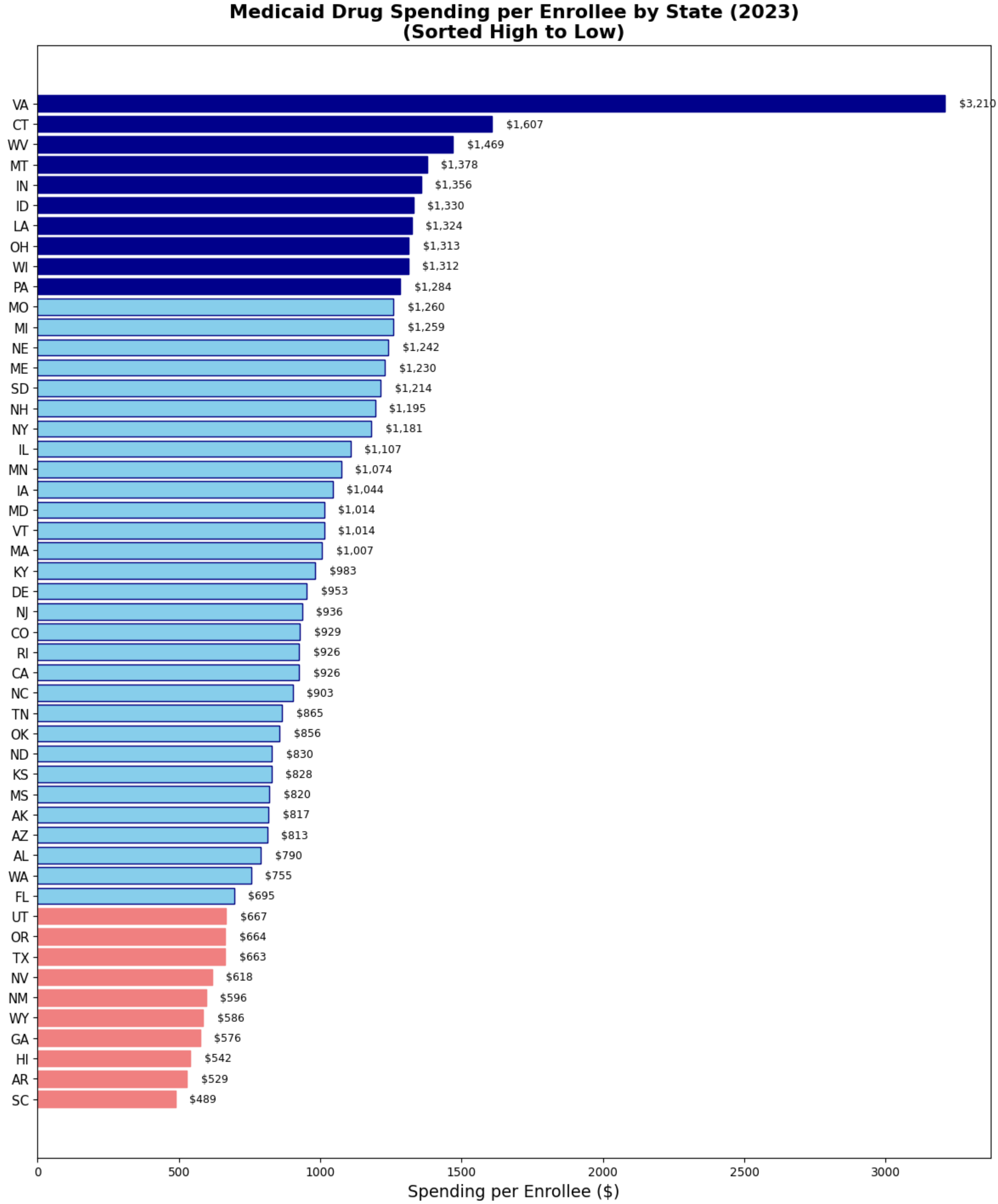
Figure 4. below shows a detailed analysis of brands accounting for 20% of Medicaid/Medicare total spending over the period of 5 years (2019–2023). Figure 4.



SOURCE: Authors' analysis of 2019 – 2023 Medicaid Part D Spending Dataset. (Run date Dec 28, 2025).

Charts in Figure 4 above show that it took 12 brands to reach 20% cumulative spending in the year 2019, 10 brands in 2020, 8 brands in 2021, 7 brands in 2022, and 6 brands in 2023. Notable trends include the consistency of the Eliquis brand over the 5-year period leading the chart as the number one most concentrated brand, the rapid rise of drugs like Ozempic which was not on the top list for the period of 2019–2020, but rose to the 6th place in 2022 and 2nd place in 2023. Trulicity also maintained a consistent presence in all the reported years with different position placements. Also, there was declining relative importance of some traditional blockbuster brands.

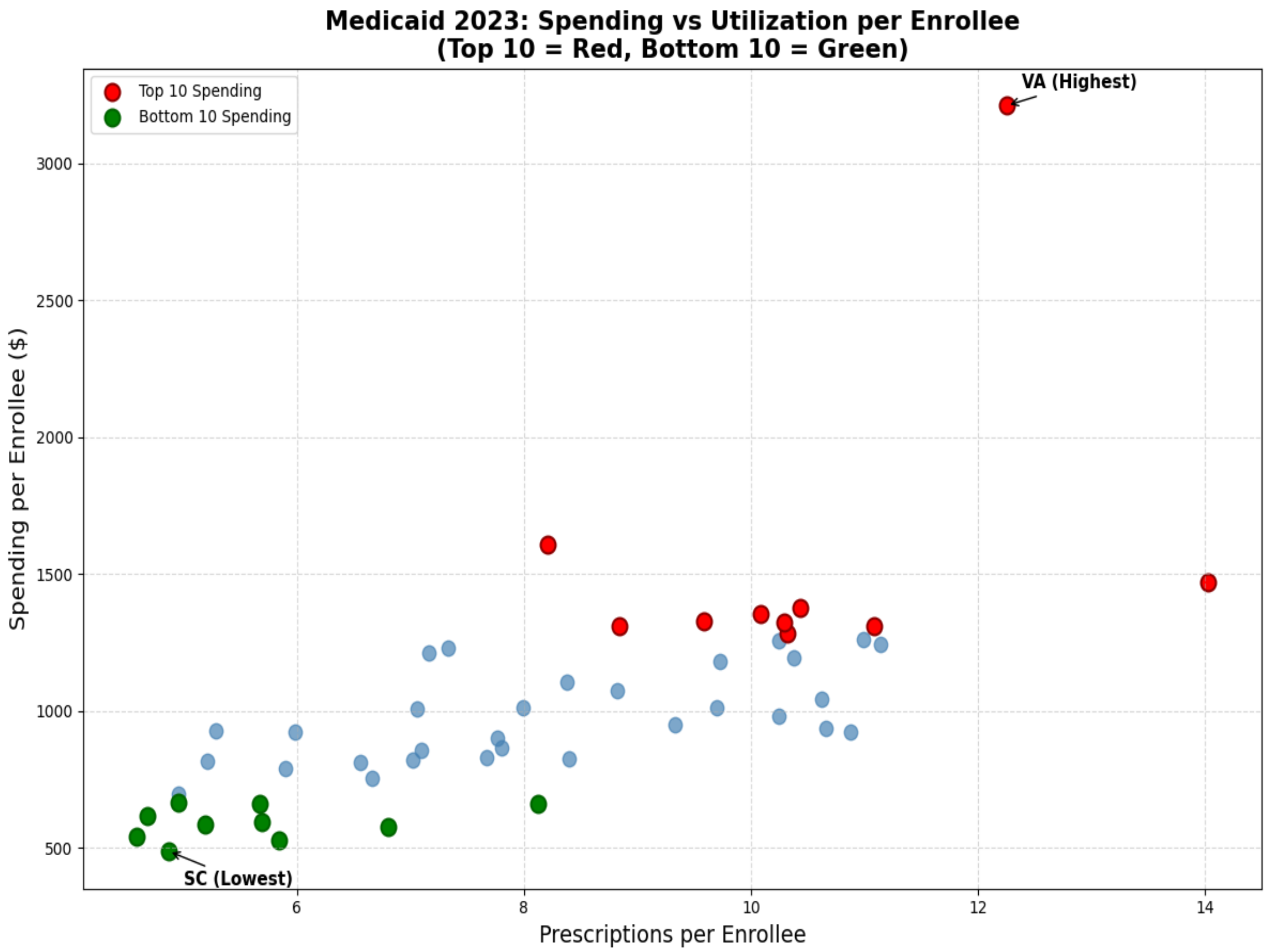
Figure 5 State-Level Variation

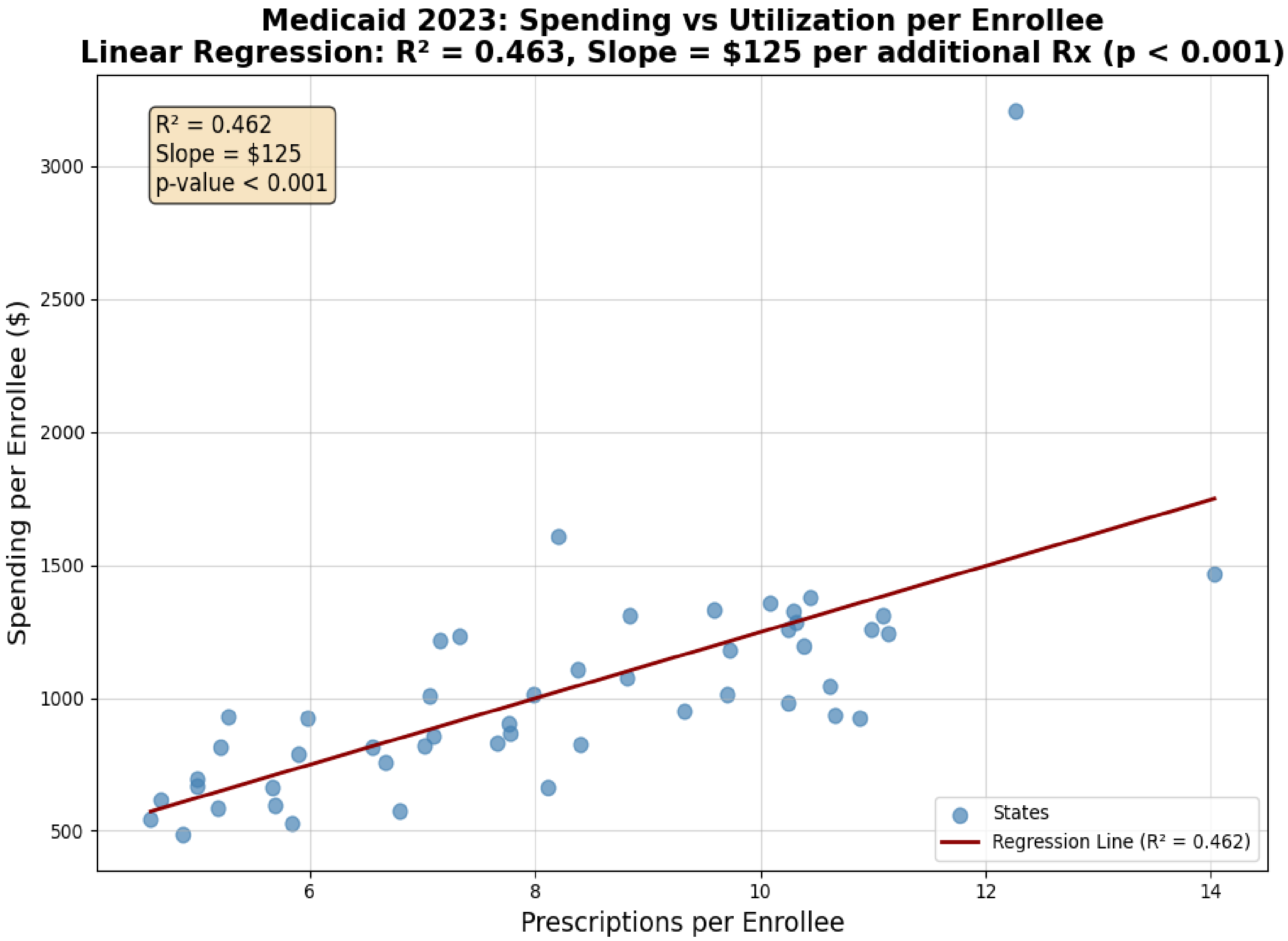


SOURCE: Authors' analysis of State Drug Utilization Data 2023 + Quarterly/State Enrollment Data. (Run date Dec 28, 2025).

In order to highlight the economic implication of the increasing total spending on the Medicaid enrollee, I analyzed the dataset of Medicaid drug spending per enrollee by merging and running analysis on datasets of 2023 state quarterly enrollee with Medicaid state drug utilization. The result shows that spending per enrollee in 2023 varied dramatically by state, ranging from $488 in South Carolina to $3,210 in Virginia with a greater than 6-fold difference. This elevated level is likely driven by Virginia having one of the highest rates of GLP-1 agonist and other diabetes/cardiovascular specialty drug prescriptions among Medicaid enrollees (Kaiser Family Foundation 2024; IQVIA Institute 2023); drugs like Ozempic, Trulicity, and Jardiance are among the top 6 high-spend prescriptions even after rebates (Figure 4 of Authors' analysis of 2019 – 2023 Medicaid Part D Spending Dataset; Run date Dec 28, 2025*)*. Virginia's Medicaid program has been relatively permissive with prior authorization for expensive specialty drugs compared to more restrictive states like California, Florida, and Texas (Medicaid and CHIP Payment and Access Commission 2023). This leads to greater access to costly therapies, driving up per-enrollee spending. Virginia also has a higher-than-average share of Medicaid enrollees who are dually eligible for Medicare and Medicaid or who qualify through disability (Kaiser Family Foundation 2023). These groups, having higher chronic disease prevalence, increase the likelihood of expensive branded medications (Centers for Medicare & Medicaid Services). Virginia also may not have aggressive negotiation when it comes to rebates, unlike states like New York and California (Government Accountability Office 2023). Also, it is possible that enrollment numbers for VA in the 2023 quarterly data were underreported or averaged differently, which would inflate per-enrollee spending (Centers for Medicare & Medicaid Services n.d.c). On average, Medicaid spent $1,020 per enrollee on drugs nationally, but there was significant state-level variation of 41%. This means that drug costs per person range widely depending on the state. This high variation of 41% among states was not fully explained by differences in prescriptions per enrollee shown in the next figure 6.

Figure 6.





SOURCE: Authors' analysis of State Drug Utilization Data 2023 + Quarterly/State Enrollment Data. (Run date Dec 28, 2025).

Figure 6 shows a scatter plot and regression of spending against prescriptions per enrollee, revealing a moderate positive correlation. This translates to the fact that states with higher utilization (prescriptions per enrollee) generally exhibit higher spending, such as Virginia, which has an average spending per enrollee of $3,210 but with higher utilization of between 12 to 13 prescriptions.

I probed further by running a simple linear regression of spending per enrollee on prescriptions per enrollee, and the outcome revealed a significant positive correlation of 0.6793 between prescriptions per enrollee and spending per enrollee ($R^2 = 0.46$, $p < 0.001$). This indicates that 46% of the variation in spending per enrollee is explainable by the number of prescriptions per enrollee. However, the remaining 54% unexplained variation is attributable to differences in drug pricing, mix, and program design across states.

Figure 6 plot reveal that high-spending States (highlighted in red) cluster at varying utilization levels, with some achieving high spending despite only moderate prescription, which could suggest reliance on very expensive brands or drug mix. Low-spending states (highlighted in green) predominantly show lower utilization (prescriptions)

5. Discussion and Modeling

5.1 Benefits of Current System

The analysis confirms that high U.S. prices generate substantial revenues for pharmaceutical companies, with public payers alone contributing approximately $552 billion in Medicaid gross spending in 2023 alone. This revenue stream is critical for funding R&D investment that benefits patients not just in the United State but worldwide.

5.2 Modeling Single Global Pricing Scenarios

According to the RAND Corporation 2024 report:

"The higher U.S. index for brand-name originator drugs (422 percent) is the primary driver of the overall difference in drug prices between the United States and other countries." (Mulcahy, Schwam, and Lovejoy 2024, vii)

"While U.S. prices across all drugs (including generics) are 278% of the OECD average, the disparity is far greater for brand-name originator drugs (422%), which dominate Medicare and Medicaid spending." (Mulcahy, Schwam, and Lovejoy 2024, vii)

My Pareto analyses in figure 4.3.1 show that a small number of high-cost brand-name originator drugs (e.g., Eliquis, Ozempic, Jardiance, Trulicity, Revlimid) account for the vast majority of the top 20% cumulative spending each year.

To better understand the impact of single global pricing scenarios on pharmaceutical industry revenue generation and cost savings to the public and insurance companies, adjustments were made using the RAND Corporation ratio of U.S. index for brand-name originator drugs of 422% and U.S. prices across all drugs (including generics) of 278% overall average difference in drug prices between the United States and other countries (Mulcahy, Schwam, and Lovejoy 2024).

To determine the percentage cut needed to unify global pricing with other members of the 33 OECD countries, this model was used: Percentage cut needed:
For 422% = $(1 - 1/R) \times 100\%$ where $R = 4.22 = (1 - 1/4.22) \times 100\% = 76\%$.
For 278% = $(1 - 1/R) \times 100\%$ where $R = 2.78 = (1 - 1/2.78) \times 100\% = 64\%$.

Hypothetical Single Global Pricing Scenario

Scenario	Price Reduction Assumption	Estimated Annual Savings	Likely Impact on Industry R&D
Moderate	64% reduction on affected spending on 2023 Medicaid total spending of $552 billion	$ 358.8 billion	Moderate reduction in revenue and profitability. This has a potential of slowing research and development
Aggressive	76% reduction on affected spending on 2023 Medicaid total spending of $552 billion	$419.52 billion	Substantial profit reduction and higher risk of fewer new drugs

Source: Authors' analysis of hypothetical single global pricing using ratios from RAND corporation's 2024 report "International Prescription Drug Price Comparisons" as adjusting factor to model price reduction scenario.

These scenarios have established that these savings would primarily benefit U.S. consumers through lower insurance premiums and out-of-pocket costs, with improved drug accessibility. These hypothetical savings align closely with recent policy efforts under the Trump administration. In May 2025, President Trump signed an Executive Order directing pharmaceutical manufacturers to adopt most-favored-nation (MFN) pricing, requiring U.S. prices to match the lowest prices paid in comparable developed nations, particularly for Medicaid and new drug launches. By December 2025, voluntary agreements had been reached with nine major manufacturers to implement MFN pricing for certain drugs, with the administration projecting significant cost reductions for public payers and patients. These developments demonstrate real-world movement toward the international alignment modeled here, though full implementation and long-term savings remain under negotiation.

5.3 Balancing Innovation Incentives and Price Reductions

It is advised that one should approach this issue with caution as there is a high risk of leaving pharmaceutical companies with little or no incentives to innovate and produce new drugs to address the increasing human demands of healthcare innovation, which may likely reverse most or all of the gains achieved by the unilateral global pricing system. This argument has been widely debated in health economics and policy literature. Pharmaceutical manufacturers often report operating margins of 15–25% on branded drugs (PhRMA 2024; IQVIA 2023), with gross margins on many blockbuster products exceeding 70–90% before R&D and marketing costs are deducted. The implied rate of return on capital (ROIC) for large U.S.-based pharmaceutical companies typically ranges from 10–20% (Statista 2025; Damodaran 2025), significantly above the average for non-pharma S&P 500 firms (~8–10%). Several studies suggest that prices could fall substantially without eliminating innovation incentives. For example, research indicates that even a 20–40% reduction in U.S. net prices would still allow returns well above the cost of capital for most companies, particularly given that R&D costs represent only 15–20% of revenues on average (Saha et al. 2021; Berndt et al. 2023). The IRA's Medicare drug price negotiations (2022–present) are projected to reduce spending by $98–266 billion over a decade with only modest estimated impacts on new drug launches (CBO 2024; Dusetzina et al. 2025). However, critics argue that sustained large cuts (e.g., 50%+) could reduce R&D investment by 10–30% over the long term, particularly for high-risk, early-stage research (Acemoglu et al. 2023; Lakdawalla et al. 2022). In summary, while the risk to innovation is real and debated, the threshold at which price reductions would meaningfully impair new drug development has not been established yet.

Conclusion

The analysis of Medicare and Medicaid drug spending from 2019 to 2023 confirms that the differential pricing system pharmaceutical pricing system provides clear net economic benefits to U.S.-based pharmaceutical companies. High U.S. prices generate substantial revenues and profits, with public payers alone contributing approximately $552 billion in gross spending in 2023 (Authors' Analyses of Medicaid part D total spending dataset). This revenue stream supports the majority of global pharmaceutical R&D investment, enabling the development of new drugs that benefit patients worldwide.

At the same time, the system imposes considerable costs on U.S. healthcare payers and consumers. Gross spending grew 54% over the five-year period, driven primarily by price increases rather than utilization (prescription) growth. Spending became increasingly concentrated in a small number of high-cost brand-name drugs, with the number of drugs required to account for 20% of total spending declining from 12 in 2019 to 6 in 2023. State-level variation in Medicaid spending per enrollee was substantial, at 41% variation, further highlighting the uneven burden at state level.

Hypothetical scenarios aligning U.S. prices with international averages suggest potential savings of $358.8 to $419.52 billion annually for public payers, which would translate into lower premiums, reduced out-of-pocket costs, and improved drug accessibility for American patients. However, such price reductions would significantly diminish pharmaceutical industry revenues and profits, carrying a high risk of slower innovation and fewer new drugs in the future.

These findings reveal a fundamental trade-off: the current international differential pricing system sustains pharmaceutical innovation through high U.S. prices but at the expense of domestic affordability and uneven state pricing. Policy that substantially narrow international price differentials would reverse this balance, offering major benefits to U.S. payers and patients while potentially jeopardizing long-term advances in medical treatment. Any future approach must carefully weigh these competing priorities to preserve incentives for innovation while improving access and cost control for Americans.

References

Acemoglu, Daron, et al. 2023. "The Impact of Drug Price Controls on Innovation." NBER Working Paper Series. Cambridge, MA: National Bureau of Economic Research.

Berndt, Ernst R., et al. 2023. "Pharmaceutical R&D Returns and Price Regulation." Health Affairs 42 (3): 123–135.

Centers for Medicare & Medicaid Services. n.d.a. "Medicare Part D Spending by Drug." Data.CMS.gov. Accessed December 30, 2025. https://data.cms.gov/summary-statistics-on-use-and-payments/medicare-medicaid-spending-by-drug/medicare-part-d-spending-by-drug.

Centers for Medicare & Medicaid Services. n.d.b. "Medicaid Spending by Drug." Data.CMS.gov. Accessed December 30, 2025. https://data.cms.gov/summary-statistics-on-use-and-payments/medicare-medicaid-spending-by-drug/medicaid-spending-by-drug.

Centers for Medicare & Medicaid Services. n.d.c. "State Drug Utilization Data." Medicaid.gov. Accessed December 30, 2025. https://www.medicaid.gov/medicaid/prescription-drugs/state-drug-utilization-data/index.html.

Centers for Medicare & Medicaid Services. n.d.d. "National Average Drug Acquisition Cost (NADAC)." Medicaid.gov. Accessed December 30, 2025. https://www.medicaid.gov/medicaid/prescription-drugs/national-average-drug-acquisition-cost-nadac/index.html.

Congressional Budget Office (CBO). 2024. "Estimated Budgetary Effects of Drug Price Negotiation under the Inflation Reduction Act." Washington, DC: Congressional Budget Office.

Dusetzina, Stacie B., et al. 2025. “Medicare Drug Price Negotiations: Early Impacts and Projections.” JAMA Health Forum 6 (1)

Government Accountability Office (GAO). 2023. Prescription Drugs: Medicare Spending on Drugs with Direct Manufacturer Payments. GAO-23-105759. Washington, DC: U.S. Government Accountability Office.

IQVIA Institute for Human Data Science. 2023. Global Use of Medicines 2023: Outlook to 2027. Durham, NC: IQVIA. https://www.iqvia.com/insights/the-iqvia-institute/reports-and-publications/reports/the-global-use-of-medicines-2023.

IQVIA Institute for Human Data Science. 2023. Global Trends in R&D 2023. Durham, NC: IQVIA. https://www.iqvia.com/insights/the-iqvia-institute/reports-and-publications/reports/global-trends-in-r-and-d-2023.

Kaiser Family Foundation (KFF). 2024. “Prescription Drug Spending in the U.S.” https://www.kff.org/health-costs/issue-brief/prescription-drug-spending-in-the-u-s/.

Lakdawalla, Darius, et al. 2022. “The Economics of Drug Pricing and Innovation.” American Economic Review: Insights 4 (3): 123–145.

Mulcahy, Andrew W., Daniel Schwam, and Susan L. Lovejoy. 2024. International Prescription Drug Price Comparisons: Estimates Using 2022 Data. RR-A788-3. Santa Monica, CA: RAND Corporation. https://www.rand.org/pubs/research_reports/RRA788-3.html.

Pharmaceutical Research and Manufacturers of America (PhRMA). 2024. “Pharmaceutical Industry Profile 2024.” Washington, DC: PhRMA. https://phrma.org.

Saha, Sajjad, et al. 2021. “Returns on Pharmaceutical R&D: Evidence from the U.S.” Journal of Health Economics 78: 102456. https://doi.org/10.1016/j.jhealeco.2021.102456.

White House. 2025. “Delivering Most-Favored-Nation Prescription Drug Pricing to American Patients.” Executive Order. May 12, 2025. https://www.whitehouse.gov/presidential-actions/2025/05/delivering-most-favored-nation-prescription-drug-pricing-to-american-patients.

White House. 2025. “Fact Sheet: President Donald J. Trump Announces Largest Developments to Date in Bringing Most-Favored-Nation Pricing to American Patients.” December 19, 2025. https://www.whitehouse.gov/fact-sheets/2025/12/fact-sheet-president-donald-j-trump-announces-largest-developments-to-date-in-bringing-most-favored-nation-pricing-to-american-patients.